

Problem Solving Quick Guide - IATF 16949 Cl. 10.2

8D - Eight Disciplines (IATF 10.2.3)

- D0 - Emergency Response: Protect customer immediately before root cause found.
- D1 - Team: Form cross-functional team with champion and problem owner.
- D2 - Problem Description: Use Is/Is Not, 5W2H to define scope clearly.
- D3 - Interim Containment: Sort, quarantine, 100% inspection at customer.
- D4 - Root Cause: Why Made (process) + Why Shipped (detection) - use 5-Why.
- D5 - Permanent CA: Action that addresses root cause, not symptom.
- D6 - Verify CA: Turn problem on and off. Get statistical evidence.
- D7 - Prevention: Update FMEA, Control Plan, SOP, training. Check similar products.
- D8 - Closure: Document lessons learned. Congratulate team.

5-Why Methodology

- Rule 1: Each answer becomes the next "Why" question.
- Rule 2: Stop when you reach a systemic or root level cause.
- Rule 3: Verify each answer - do not assume.
- Rule 4: Address both Why Made AND Why Shipped separately.
- Rule 5: The root cause must be fixable with a systemic action.

Fishbone (Ishikawa) - 6M Framework

- Man: Skill, training, fatigue, operator error, competency.
- Machine: Wear, breakdown, calibration, tooling, PM status.
- Material: Raw material variation, handling, supplier quality.
- Method: SOP not followed, setup error, process parameters.
- Measurement: Gauge R&R, instrument calibration, inspector bias.
- Mother Nature: Temperature, humidity, vibration, lighting, dust.

CAPA Effectiveness Verification (IATF 10.2.3)

- After implementing CA - monitor for minimum 3 months (or 1 production batch).
- Check: Has the same failure mode recurred? (Zero recurrence = effective)
- Check: Is the control still in place? (SOP, poka-yoke, gauge)
- Document effectiveness result in CAPA register.
- If not effective - re-open CAPA with improved action.

Common Audit Findings - Problem Solving

- Major NC: No 8D raised for customer complaints within required timeline.
- Major NC: Root cause identified is a symptom - not the systemic root cause.
- Minor NC: 8D D7 prevention - FMEA/Control Plan not updated.
- Minor NC: CAPA effectiveness not verified after corrective action closure.
- Best Practice: Lessons Learned linked to DFMEA, PFMEA and Control Plan.

- Best Practice: Problem Solving training for all Quality and Production staff.